

Supplementary material for “Walls or cascade? A pre-registered test of wall dissipation in decaying magnetohydrodynamic turbulence”

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Equations, sections, figures and tables numbered without an S are those of the paper. Every number below that comes from a run is printed, with its source, by the script that produces the paper’s numbers (`paper/tg_mhd/numbers.py` in the repository).

S1. VERIFICATION OF THE METHOD

The parity wall. Three tests in the repository check Eq. (6) (`validation/mhd_parity_wall.cpp`). First, with free slip and either parity, the box of N nodes reproduces the periodic 2π box of $2(N-1)$ nodes to 10^{-14} , node for node and transient included. Second, $\mathbf{B} \cdot \mathbf{n}$ on the wall nodes stays at 10^{-32} . Third, in the no-slip box the exact decay of the conducting-wall eigenmode $\mathbf{B} = B_0(\sin x \cos y, -\cos x \sin y, 0)$, which leaves the fluid at rest, converges at second order (order 2.00 over $N = 17, 33, 65$).

The two implementations. The C++20/Kokkos code and the CUDA code share no source. Their host builds agree to every printed digit over 815 steps at $N = 65$, $\text{Re} = 200$, with free and no slip, and the CUDA build on the device agrees with the Kokkos reference to within 5.5×10^{-16} of each diagnostic’s scale, in four verification jobs, each with free and no slip. The runs behind the paper’s numbers took 34.1 GPU-hours: the ladder, its two controls, and a second round with the top rung, the insulating pair, and reruns of the lower rungs that also write the viscous and Ohmic parts of the profile. The reruns that wrote the fields of Sec. IV F took 1.5 more, and the first ladder, run with the estimator of Sec. S3, 8.2.

Run C2 of Pouquet et al. The free-slip box A at $\text{Re} = 1000$ is run C2 of Ref. 1. Its minimum ratio of magnetic to kinetic energy is $E_M/E_V = 0.3695$ and 0.3693 at $N = 384$ and 512 , reached at $t = 7.3$, against their 0.35 , and the first maximum of Ω_M/Ω_V after its initial fall is 2.154 and 2.159 , against their “2.”. Our ratio is converged to four figures between the two grids and lies 5.5% above theirs, which is quoted to two figures.

We attribute the gap to the resolution of C2. It is the coarsest run of Ref. 1: pseudo-spectral on an equivalent grid of 128^3 over the period 2π , i.e. 64 points across our box. Its Table 1 lists $\delta k_{\max} = 0.8$, where δ is the logarithmic decrement of a fit $E_T(k) \propto k^{-n} e^{-2\delta k}$ to the energy

spectrum and $k_{\max} = N/3$. That is below $\delta k_{\max} \approx 1$, at which the authors stop an ideal computation, and below the $\delta k_{\max} \geq 2$ by which they call their 2048^3 runs well resolved; their other two 128^3 runs have 1.9 and 2.1, and C2 has no companion at higher resolution. Under-resolution moves our own ratio the same way: 0.357 at $N = 97$, reached at $t = 9.8$, and 0.329 at $N = 65$, still falling at $t = 10$. The two codes under-resolve by different mechanisms, so the common direction is suggestive rather than conclusive.

S2. THE RECORD OF THE PROTOCOL

The two outcomes of Sec. I were stated, qualitatively, on 25 September 2026 in a plan that is not in the repository, and nothing there dates them. The rules of Sec. IID were committed with the job scripts. A commit’s date is written by its author, so Table S1 also gives the time at which each commit reached GitHub, from GitHub’s server-side record of the pushes to the repository. A copy of that record, as GitHub’s public interface returned it on 2 October 2026, is kept in the repository (`paper/tg_mhd/provenance/`) with the request that fetched it; GitHub returns it to anyone for about three months after a push. Each rule was on GitHub within 80s of being committed. The estimator was corrected after the first ladder had failed its gate (Sec. S3); the observable it estimates and the gate did not change. The Mach control, and the decision rule and the precision control for the top rung, were fixed after the lower rungs had been run, and seen, and before the runs they judge. Before 30 September the ladder’s script asked only that $B-A$ exceed its band and trend monotonically over at least three rungs; the decision rule of Sec. IID, which is the one applied in the paper, replaced that with a test at the top rung, fixed with the job script that ran it.

S3. THE ESTIMATOR OF THE WALL SHARE

Counting whole node layers, every node with $kh < m\delta$, measures a layer of thickness $(\lceil m\delta/h \rceil - \frac{1}{2})h$ rather than $m\delta$, because the nodes with $d = kh$ fill exactly the slab between $(k - \frac{1}{2})h$ and $(k + \frac{1}{2})h$ (Sec. II): 0.91δ at $N = 384$ and 1.07δ at $N = 512$ for $\text{Re} = 1000$. Our first ladder counted layers. At every rung of every box with a turbulent peak, eight in all, it failed the gate: the worst

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TABLE S1. When each rule was fixed: the commit, its time by the author’s clock, and the time at which it reached GitHub by GitHub’s push record, on the same day. All times UTC, 2026.

rule	commit	committed (UTC)	pushed (UTC)	lag (s)
observable, peak rule, gate	663929c	27 Sep, 11:57:38	11:58:24	46
estimator of f_m corrected	9f65706	28 Sep, 11:55:32	11:56:52	80
Mach control	cd07f45	29 Sep, 06:22:12	06:22:15	3
decision rule, FP64 control (Re = 2000)	d061ee8	30 Sep, 20:03:29	20:03:34	5

of f_1 , f_2 and f_4 differed between the two grids by 6.5–24.6%, while ε at the peak agreed to 1%. Moved to the exact band, by interpolating each grid’s three samples, the same series pass, the worst difference becoming 2.9%. Swept across $N = 33$ to 49 at fixed physics (Re = 100), the count makes f_4 jump between 0.52 and 0.60 each time $4\delta/h$ crosses a node, while Eq. (7) keeps it within 0.561–0.566. Any layer, shell or threshold observable on a lattice can do this.

S4. RESOLUTION GATE AND CONTROLS

Table S2 gives the resolution gate and the controls of Sec. III rung by rung.

S5. THE THICKNESS OF THE LAYER

Table S3 repeats the readings of Table III for the layers 2δ and 4δ thick. The bands are the two grid differences of each layer’s share, added, as for δ ; the largest is 0.0056 at the peak and 0.0016 over a time window. The thicker layers read as the thinnest does. B–A is positive in every reading at every rung, from 0.019 to 0.115. From Re = 250 to 2000 it changes by +2% within 2δ and –20% within 4δ over $t = 2$ –6, against –3% within δ ; falls by 12–42% and 23–35% over the windows that reach further into the decay, against 18–43%; and falls by 68% and 67% at the peak, against 51%.

S6. INSULATING WALLS

The pair A’, B’ was meant to show whether the result needs a conducting wall. It cannot. A’ passes the gate (worst 0.8%), and 88% of the dissipation within δ of its walls is Ohmic ($F_1 = 0.31$ and 0.27 at Re = 500 and 1000). B’ has no turbulent peak by $t = 10$ at either Re. It puts more of its dissipation within δ of the walls than any other box, $F_1 = 0.63$ and 0.69, and these agree between its two grids to 1.4% and 1.7%. So B’–A’ (+0.32 and +0.42 over $t = 2$ –10) compares a box that has not become turbulent with one that has, and under a different initial field from B–A.

B’ is also the only box in which the field crosses a no-slip wall, the configuration of a Hartmann layer.^{2,3} With the field normal to the wall, B_n , and a tangential velocity U outside the layer, the tangential momentum and induction equations in the layer are, in Alfvén units, $\nu u'' + B_n b' = 0$ and $B_n u' + \eta b'' = 0$, primes denoting derivatives along the normal. They give $u = U(1 - e^{-z/\delta_H})$ with $\delta_H = \sqrt{\nu\eta/|B_n|}$, and a dissipation per unit area of

$$\int_0^\infty (\nu u'^2 + \eta b'^2) dz = U^2 |B_n| \sqrt{\nu/\eta}, \quad (\text{S1})$$

half of it viscous and half Ohmic. At Pm = 1, $\delta_H = 1/(\text{Re}|B_n|)$ has the scaling of Kato’s layer, and Eq. (S1) does not depend on Re. The field $\mathbf{B}_I$ of Eq. (5) is normal to every face at $t = 0$, with $|B_n|$ up to $2b_0 = 1.15$ on the two faces normal to z and up to b_0 on the other four. There a Hartmann layer would be $0.14h$ and $0.28h$ thick at Re = 1000, and $0.21h$ and $0.42h$ at Re = 500, on the finer grids. These are estimates from the initial field: we have not measured the layers of B’, and its integrated share agrees between grids that could not resolve them. We draw no conclusion from B’ or from B’–A’.

S7. THE FIELDS AT A LATER TIME

Figures S1 and S2 repeat Figs. 4 and 5 at $t = 4.6$, by the later maxima of the dissipation. By then the bulk of both MHD boxes is more finely structured and A’s near-wall plane has quietened, but the contrast of Sec. IV F remains: in the plane 0.58δ above the bottom wall, B’s viscous dissipation is 6.8 times A’s and its Ohmic 0.68 of A’s. In three dimensions both MHD boxes have broken up into finer sheets, and the viscous dissipation of the no-slip box forms ring-like patterns on its faces. The box without a field dissipates 3.3 times its box mean in its near-wall plane, about 3.2 times B’s relative level.

¹A. Pouquet, E. Lee, M. E. Brachet, P. D. Mininni, and D. Rosenberg, “The dynamics of unforced turbulence at high Reynolds number for Taylor–Green vortices generalized to MHD,” *Geophys. Astrophys. Fluid Dyn.* **104**, 115–134 (2010).

²J. Sommeria and R. Moreau, “Why, how, and when, MHD turbulence becomes two-dimensional,” *J. Fluid Mech.* **118**, 507–518 (1982).

³B. Knaepen and R. Moreau, “Magnetohydrodynamic turbulence at low magnetic Reynolds number,” *Annu. Rev. Fluid Mech.* **40**, 25–45 (2008).

TABLE S2. Resolution gate and controls. Relative differences of f_1 , f_2 and f_4 at the turbulent peak between the two grids of each rung, the band of B–A (the two grid differences of f_1 , added), and the shift of B–A in the control run: Mach, u_0 halved; FP64, double precision at $N = 512$.

Re	N	A: $\Delta f_1/\Delta f_2/\Delta f_4$ (%)	B: $\Delta f_1/\Delta f_2/\Delta f_4$ (%)	band	control shift
250	192, 256	0.7/0.2/0.1	1.0/0.7/0.2	0.0020	–
500	256, 384	0.8/0.5/1.0	2.1/1.5/0.7	0.0030	Mach, -0.0014
1000	384, 512	0.7/0.7/0.3	0.2/0.1/0.2	0.0004	Mach, $< 10^{-4}$
2000	512, 640	3.0/1.3/0.8	0.5/0.4/0.3	0.0021	FP64, $< 10^{-4}$

TABLE S3. B–A in the share of the dissipation within $m\delta$ of the walls, at the turbulent peak and integrated over the time windows of Table III, on the finer grid of each rung. The last column is the change from Re = 250 to 2000.

reading	m	250	500	1000	2000	change (%)
at the peak (pre-registered)	1	0.073	0.065	0.063	0.036	–51
	2	0.075	0.076	0.065	0.024	–68
	4	0.058	0.050	0.064	0.019	–67
integrated, $t = 2-6$	1	0.049	0.048	0.051	0.048	–3
	2	0.047	0.044	0.051	0.048	+2
	4	0.060	0.049	0.050	0.048	–20
integrated, $t = 2-10$	1	0.059	0.055	0.054	0.048	–18
	2	0.055	0.056	0.057	0.048	–12
	4	0.063	0.061	0.059	0.049	–23
integrated, $t = 3-10$	1	0.066	0.059	0.055	0.046	–29
	2	0.066	0.062	0.058	0.044	–34
	4	0.066	0.070	0.063	0.044	–32
integrated, $t = 4-10$	1	0.082	0.072	0.064	0.050	–39
	2	0.084	0.080	0.070	0.049	–42
	4	0.078	0.088	0.077	0.050	–35
integrated, $t = 5-10$	1	0.095	0.085	0.072	0.054	–43
	2	0.093	0.100	0.084	0.056	–40
	4	0.087	0.115	0.098	0.061	–30

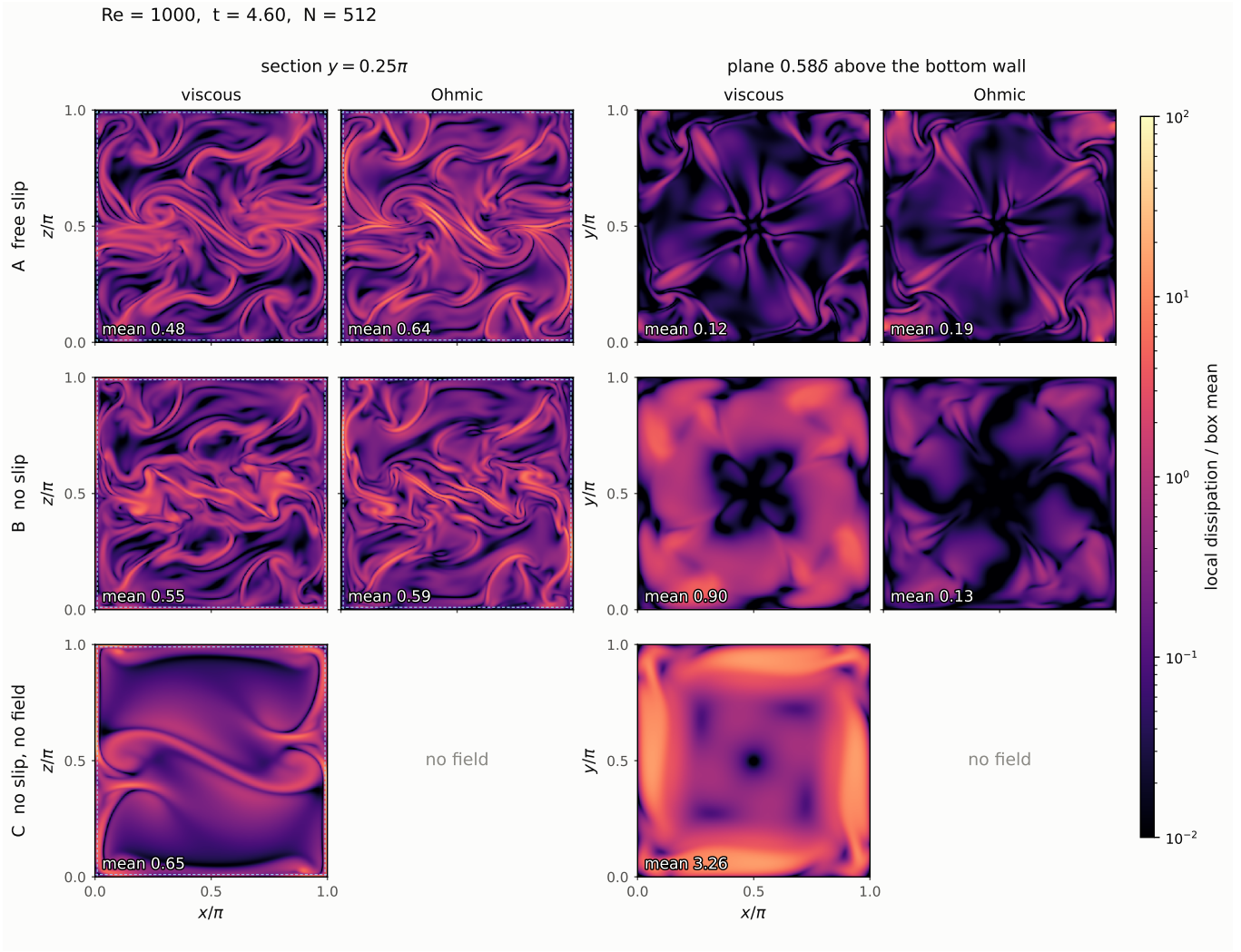


FIG. S1. As Fig. 4, at $t = 4.6$, by the later maxima of the dissipation.

$Re = 1000$, $t = 4.60$, $N = 512$ (block maximum over 2^3)

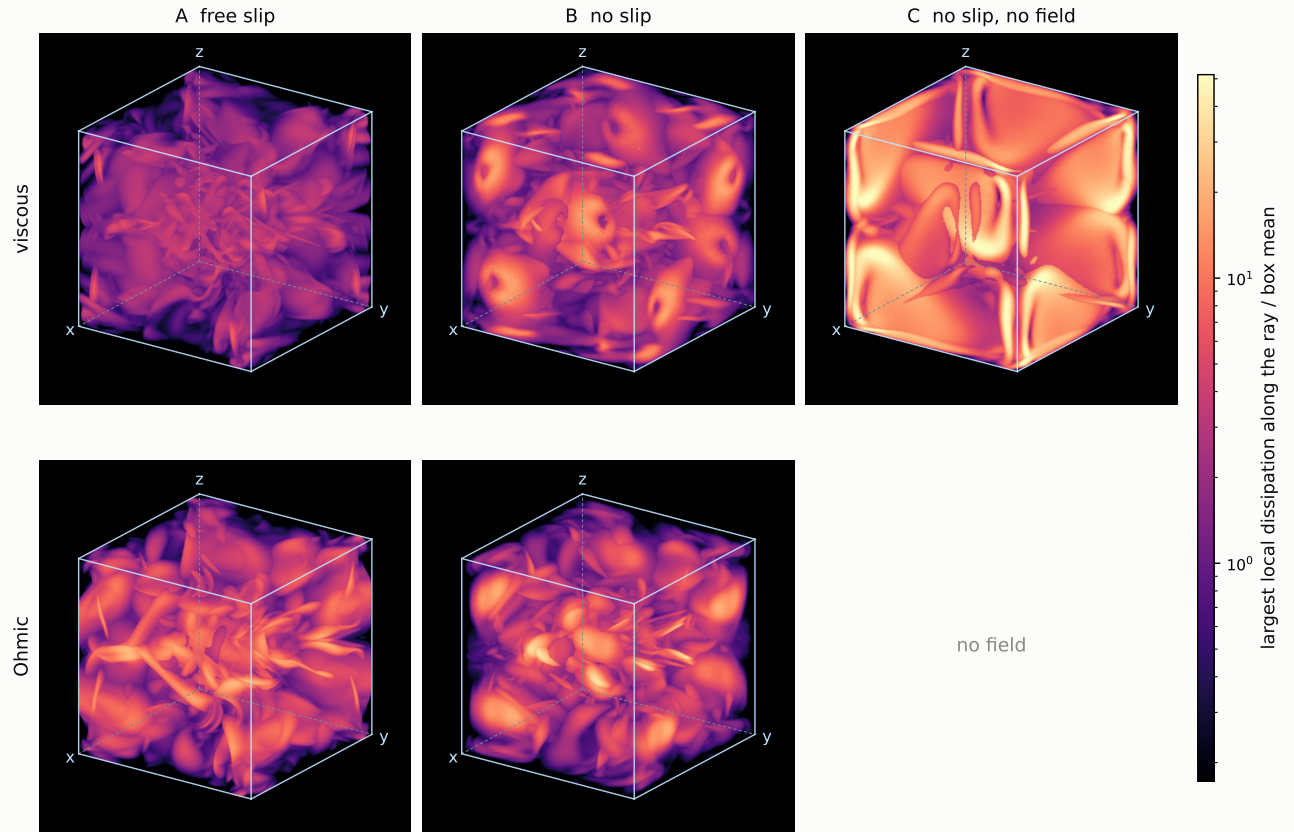


FIG. S2. As Fig. 5, at $t = 4.6$, on the same colour scale.